

MLCC 2026

Lab 3 – Logistic Regression

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Logistic Regression

ERM $\min_{w \in \mathbb{R}^D} \frac{1}{n} \sum_{i=1}^n L(y_i, f_w(x_i)) + \lambda \|w\|^2$

Loss $L(y_i, f_w(x_i)) = \log(1 + e^{-y_i f_w(x_i)})$

Despite the name we are solving a **classification task**!

Logistic Regression

Gradient descent

$$w_t = w_{t-1} - \gamma \left(\frac{1}{n} \sum_{i=1}^n x_i \frac{-y_i}{1 + e^{y_i x_i^T w_{t-1}}} + 2\lambda w_{t-1} \right)$$

Logistic Regression

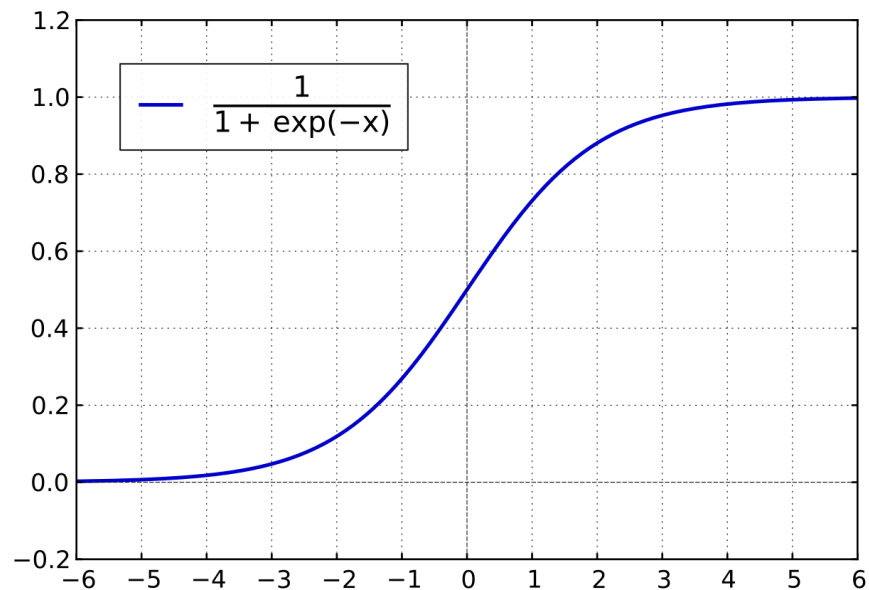
Stochastic gradient descent

$$w_t = w_{t-1} - \gamma_t \left(x_i \frac{-y_i}{1 + e^{y_i x_i^T w_{t-1}}} + 2\lambda w_{t-1} \right)$$

Logistic Regression

Confidence

$$\frac{1}{1 + e^{-yf_w(x)}} = \frac{1}{1 + e^{-y w^T x}} \xrightarrow{y=1} \frac{1}{1 + e^{-w^T x}} = p(y = 1|x)$$



Logistic Regression

Goals:

- Implement Logistic regression with Gradient Descent
- Investigate the confidence
- Extend to Stochastic GD